

IK 503: Cognitive Psychology and the Indian Thought System

Course Code	IK 503
Course Name	Cognitive Psychology and the Indian Thought System
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BTech/MTech/MS/MSc/MA/Ph.D.
Mutual Exclusion	None
Approval	50th BoA (Jul 2023); 42nd Senate (Jan 2024)

Course Contents

- **Evolution, Mind, and Brain:** Nervous system - anatomy and physiology; Functional neuroanatomy; Tools for investigation – electrophysiology, imaging, and others; how the brain creates mind?; Translation to behavior — emotion/cognition/decision making; mental representations and processing; dissociations and associations. (5 Hours)
- **The Indian Knowledge System:** Six Schools of philosophy; Buddhism; Bhagavad Gita; Mapping with the Neuroscientific/psychological understanding from Unit 1; Mental health; cognition in Samkhya and yoga; the body – mind – intellect – consciousness complex; consciousness; panca – kosa – a five layered existence; four states of existence; driving issues in consciousness studies; the tri – guna system; cognitive training hypothesis in yoga; psychological effects of yoga/meditation with clinical and nonclinical populations; Extraordinary cognition hypothesis via eightfold path described in the Yogasutras. Relative versus absolute reality hypothesis. (7 hours)
- **Perception and Attention:** Introduction to perception; visual perception; structure of visual system; top-down (context effects) and bottom-up (from features to objects) processing; visual recognition; interactive nature of perception; nature and roles of attention; failures of selection; successes of selection; information processing theories of attention; electrophysiology and human attention; functional neuroimaging and transcranial magnetic stimulation. (7 Hours)
- **Representation, Encoding, and Retrieval of Knowledge in Long-Term Memory:** Role of knowledge in cognition; representations and their formats; representation to category knowledge; structures in category knowledge; category domains and organization; nature of long-term memories; encoding; retrieval; encoding with difficulty to recall; non-declarative memory systems. (7 Hours)
- **Working Memory and Executive Processes:** Introduction to working memory; from primary memory to working memory; working memory models; person-to-person variation; dopamine's role; frontal lobe connection; frontal damage and the frontal hypothesis; executive attention; switching attention; inhibition of response; sequencing; monitoring. (6 Hours)
- **Emotion, Cognition, Decision-making, and Problem Solving:** Defining emotion; manipulating and measuring emotion; emotional learning; acquiring evaluations; emotion and declarative memory; emotion, attention, and perception; nature of a decision; rational decision making; neural bases of expected utility calculations; human decision making and the expected utility model; complex, uncertain decision making; nature of problem solving; analogical reasoning; inductive reasoning; deductive reasoning. (7 Hours)
- **Language, Motor Cognition, and Mental Simulation:** Nature of language; processes of language comprehension; processes of language production; language, thought, and bilingualism; nature of motor cognition; mental simulation and the motor system; imitation; biological motion. (5 Hours)

Text Books

1. Smith, E. E., & Kosslyn, S. M., **Cognitive Psychology: Mind and Brain**, Pearson, 2013.
2. Eysenck, M. W., & Keane, M. T., **Cognitive Psychology, A Student's Handbook**, 8th Edition, Hove: Psychology Press, 2020.

References

1. Ward, J., **The Student's Guide to Cognitive Neuroscience**, 3rd Edition, Hove: Psychology Press, 2015
2. Anderson, J. R., **Cognitive Psychology and Its Implications**, 9th Edition, Worth Publishers, 2020.
3. Sedlmeier P. & Srinivas K., How Do Theories of Cognition and Consciousness in Ancient Indian Thought Systems Relate to Current Western Theorizing and Research? **Front Psychol.** 2016 Mar 15;7:343. doi: 10.3389/fpsyg.2016.00343, 2016.